

Abstract

Objective: To determine how seizure-detector sensitivity and false alarm rate affect observed regression to the mean and apparent placebo response in simulated epilepsy randomized controlled trials.

Methods: Using a seizure-diary simulator, we generated 100,000 potential participants with a two-month baseline and three-month test period. Separate sweeps varied sensitivity from 10% to 100% and false alarm rate from zero to one alarm/day. Expected false alarms were subtracted from monthly counts. Eligibility used monthly thresholds without a seizure-free-interval criterion. Sensitivity analyses retained the former 25-day rule, held cohort membership fixed, and used additive expected-alarm eligibility thresholds.

Results: With perfect detection, 37.2% of eligible participants met the observed regression-to-the-mean definition and median placebo change was 14.3% (95% confidence interval 13.3% to 14.8%). At 10% sensitivity, these values were 64.4% and 48.1%; at one false alarm/day, they were 53.6% and 31.2%. The 25-day rule preserved the overall trends. Holding the cohort fixed reversed the rising observed regression-to-the-mean pattern. Additive thresholds produced identical outcomes.

Significance: Detector errors changed eligibility and apparent placebo response despite expected-alarm correction. The increasing observed regression-to-the-mean pattern depended on reselection and should not be interpreted as an isolated measurement-error effect within a fixed cohort. Trial simulations should specify detector performance, correction rules, and eligibility together.

Keywords: seizure detection; regression to the mean; clinical trials; false alarms; simulation

Introduction

Automated seizure detection devices are evaluated by sensitivity, false alarm rate (FAR), detection latency, and usability. Guidelines and meta-analytic studies show that performance varies across seizure types and recording settings.¹⁻³ These metrics are commonly interpreted for patient safety, caregiver notification, or diary accuracy. They can also affect eligibility and outcome calculations in randomized controlled trials (RCTs).

Epilepsy trials are vulnerable to apparent placebo response for several reasons. Seizure frequency varies over time, affecting outcomes.⁴⁻¹⁰ Placebo response includes regression to the mean

(RTM).^{8,11} Prior work showed that eligibility rules can change RTM and placebo response.⁴ Natural seizure-frequency variability can reproduce many features of placebo response without a psychobiological placebo effect.^{5–8,12}

Seizure detection devices change measured counts before trial eligibility and outcomes are calculated. We investigated how sensitivity and FAR alter a study-specific observed-RTM pattern and apparent placebo response in simulated epilepsy RCTs.

Methods

Simulations used Python and CHOCOLATES, an open-source seizure-diary simulator developed to reproduce empirical diary properties, including heterogeneous seizure rates, the relationship between mean frequency and variability, clustering, and cycles.¹² We used daily sampling and 30-day months. The implemented daily model does not represent 12-hour or 24-hour within-day cycles; software versions and simulator settings are reported in the Appendix.

Each simulated placebo-arm participant had a two-month baseline and three-month test period. No active treatment effect was added. Apparent improvement can arise through natural variability and RTM without an intervention.^{4,12,13} Primary eligibility required a mean baseline frequency of at least four seizures/month and at least three in each baseline month. These thresholds adapt the Krauss cenobamate trial, which required at least eight seizures over eight weeks, at least three in each four-week segment, and no seizure-free interval longer than 25 days. The earlier Chung trial required at least three seizures/month and no consecutive 21-day seizure-free period.^{14,15} We used 30-day simulated months rather than four-week segments and omitted the interval criterion from the main analysis. The Appendix evaluates the 25-day rule separately.

Each latent seizure had the specified probability of detection. Independent Poisson false alarms were added to daily counts. For FAR conditions, the known expected number of false alarms was

subtracted from each 30-day count, and negative corrected counts were set to zero.³ This aggregate correction does not identify individual false alarms or guarantee unbiased counts after flooring. Equations and alternative rules are in the Appendix.

Two sweeps varied sensitivity from 10% to 100% in 10% increments at zero FAR, and FAR from zero to one alarm/day in 0.1-alarm/day increments at full sensitivity. A common cohort of 100,000 latent participants was reused with nested detector realizations. Seeds were 20260813 for latent histories, 20260814 for detection, and 20260815 for false alarms. Each participant's trial and separate 36-month reference diary shared the sampled rate, cluster status, and cycle parameters but had separate random count realizations. The effective long-term frequency was the mean monthly count in that reference diary, not the sampled rate parameter. It incorporates realized variability and was scaled by sensitivity for corrected-count comparisons. Observed RTM required baseline above this reference and test-period frequency strictly closer to it. Exact integer-scaled comparisons excluded equal-distance ties. This indicator describes trajectories, not causal RTM subtypes. Apparent placebo response was median percentage change (MPC) from baseline to test, with positive values indicating seizure-frequency reduction. Binomial-order-statistic 95% confidence intervals describe uncertainty in the median, not variation among individuals. No hypothesis tests were performed.

A sensitivity analysis held membership fixed to participants eligible under perfect detection and compared them with cohorts reselected at each condition using the same latent histories and detector counts. Observed RTM used the full selected denominator; MPC excluded undefined percentage changes at zero measured baseline, with those counts reported separately. Another analysis applied additive expected-alarm thresholds to raw baseline counts while retaining

corrected outcomes. Full methods, equations, all conditions, and selection diagnostics are in the Appendix.

No human participants, identifiable private information, or animals were used. The canonical script, fixed seeds, executed Appendix, aggregate outputs, run metadata, and validation checks are available in the open-source GitHub repository.

Results

At zero FAR, lower sensitivity increased observed RTM and placebo MPC (Figure 1A,C). With perfect detection, 35,393 of 100,000 participants were eligible, observed RTM was 37.2%, and MPC was 14.3% (95% CI 13.3% to 14.8%). At 50% sensitivity, 17,865 were eligible, observed RTM was 44.5%, and MPC was 25.0% (95% CI 24.6% to 25.9%). At 10% sensitivity, 2,180 were eligible, observed RTM was 64.4%, and MPC was 48.1% (95% CI 45.8% to 50.0%).

At full sensitivity, increasing FAR increased observed RTM and placebo MPC after expected-monthly-alarm correction (Figure 1B,D). At FAR 0.1/day, 35,453 participants were eligible, observed RTM was 40.8%, and MPC was 17.9% (95% CI 17.1% to 18.5%). At FAR 0.5/day, 35,508 participants were eligible, observed RTM was 49.4%, and MPC was 27.3% (95% CI 26.7% to 27.8%). At FAR 1/day, 35,134 participants were eligible, observed RTM was 53.6%, and MPC was 31.2% (95% CI 30.4% to 33.3%).

Retaining the original 25-day rule preserved the overall trends. Across 21 conditions, the maximum absolute differences were 2.28 percentage points for eligibility, 2.49 for observed RTM, and 2.29 for MPC. Additive expected-alarm thresholds selected identical participants and produced identical outcomes at every FAR. Full results are in the Appendix.

Holding the 35,393-participant perfect-detector cohort fixed changed the pattern. Observed RTM fell from 37.2% to 29.3% at 10% sensitivity and 34.7% at FAR one/day. Fixed-cohort MPC was

33.3% and 10.4%, respectively, but was undefined for 6,182 and 538 participants with zero measured baseline. The Appendix reports the changing MPC denominators.

Discussion

Device performance should be treated as a trial-design parameter. Measured baseline counts determine eligibility and form the denominator of percentage-change outcomes.⁴ Detector errors change both pathways, as illustrated in Figure 2. The rising primary observed-RTM pattern reversed when the perfect-detector cohort was held fixed. Thus that increase depended on reselection in these sweeps and was not a universal consequence of adding measurement error within an unchanged cohort. This comparison does not partition RTM into additive causal components. Fixed-cohort MPC also needs caution because zero-baseline participants have undefined percentage change, particularly at low sensitivity.

Earlier epilepsy RTM simulations showed that eligibility criteria can change observed RTM and placebo response.⁴ The present analysis examines how detector errors interact with that selection. Trials with the same biological treatment effect and nominal count thresholds may show different apparent placebo responses when detector performance or correction rules differ. Moving expected alarms from counts to additive eligibility thresholds is algebraically equivalent under our positive monthly thresholds. It simplifies expression of the rule but cannot reduce residual false-alarm noise.

A historical framework described RTM type 1 as return after a temporary unusual circumstance and RTM type 2 as return associated with ordinary natural variability.⁸ Measurement error can be considered a third contributor, but this simulation does not identify an isolated participant-level "RTM Type 3" category. Detector error changes cohort selection and the frequency of the overall observed-RTM indicator. We interpret the results as a measurement-related contribution to

observed RTM and apparent placebo response, not evidence that RTM Type 3 alone was directly observed.

Objective seizure detection could reduce under-reporting and may improve statistical power when more complete seizure counts are available.^{3,16} Device operating characteristics should be incorporated into trial simulations before randomized controlled trial designs are finalized. Before trial initiation, investigators should specify how sensitivity, FAR, correction, device non-wear, and missing data enter eligibility, baseline counts, and outcome counts. Device validation should also report performance in seizure types and settings that match the planned trial population.^{1,2}

Published studies provide context for the examined performance ranges. A multimodal wrist detector reported 94.6% sensitivity with 0.20 false alarms/day. An Apple Watch algorithm reported 100% sensitivity with 0.05/day in epilepsy monitoring units and 0.13/day in ambulatory monitoring. A connected shirt reported 84.8% sensitivity with a median FAR of 0.55/day.^{17–19} These estimates concern convulsive seizures and differ in setting, sample size, and FAR summary. The shirt study used cross-validation, and Apple Watch sensitivity estimates came from small seizure samples. Our separate sweeps do not predict outcomes for these devices at their paired sensitivity/FAR operating characteristics.

CHOCOLATES simplifies real-world seizure occurrence, and its empirical calibration diaries can contain reporting errors.¹² Treating generated events as biological truth is therefore an idealization. Daily sampling omits 12-hour and 24-hour within-day cycles, so the study cannot assess their interaction with time-dependent detector performance. Sensitivity and FAR were constant across participants and time; real detectors can have seizure-type dependence, correlated errors, non-wear, and signal loss. Expected-alarm subtraction is not event-level correction. The Appendix compares correction and interval strategies, but does not isolate the effect of zero-flooring alone.

Oracle removal eliminates the FAR effect by construction and is unavailable without alarm identities. The effective long-term reference is a finite 36-month realization. Confidence intervals quantify Monte Carlo precision under the assumed model, not uncertainty in clinical generalizability.

Seizure detection devices may make epilepsy trials more objective, but they also make trial statistics depend nonlinearly on detector performance, correction, and eligibility. Observed RTM and apparent placebo response should therefore be evaluated jointly with sensitivity and FAR whenever device-derived counts are used for epilepsy trial eligibility or endpoints.

Key points

- In these simulations, detector errors changed trial eligibility and apparent placebo response.
- The rising observed-RTM pattern depended on reselecting the cohort as detector performance changed.
- Subtracting expected false alarms left random count error that affected trial outcomes.
- The observed-RTM indicator describes trajectories without separating causal mechanisms.

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Ethical publication statement

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Data and code availability

The canonical script, unchanged simulator source, fixed seeds, executed main-paper and Appendix notebooks, aggregate results, tests, and validation metadata are available at <https://github.com/GoldenholzLab/RTM3>. The notebooks generate the manuscript figures through shared functions.

Author contributions

Daniel M. Goldenholz conceived the analysis, wrote and revised the code, interpreted results, and drafted the manuscript. Rohan Bhansali, Shira R. Goldenholz, Ted J. Kaptchuk, and Brandon Westover contributed to interpretation and manuscript revision.

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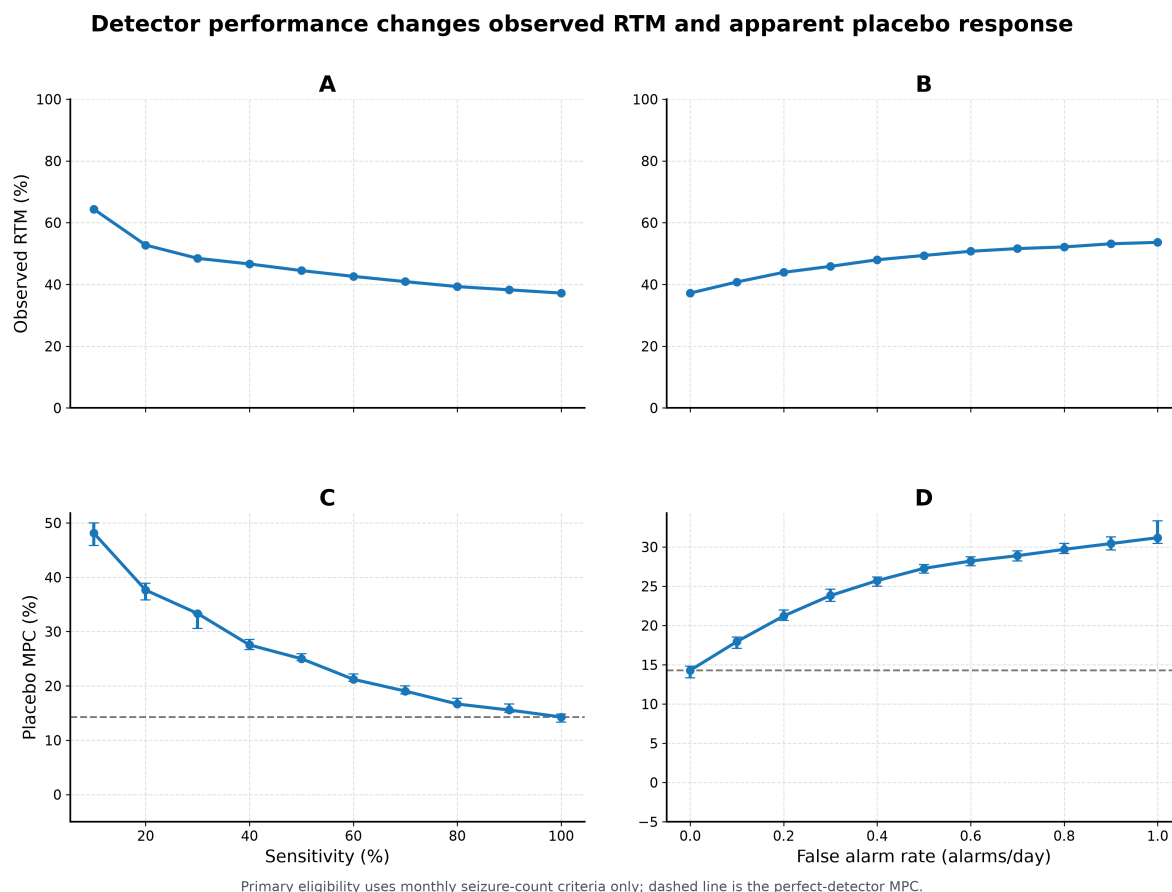


Figure 1. Detector performance, observed regression to the mean, and apparent placebo response. A,C: sensitivity sweep at zero false alarm rate. B,D: false-alarm-rate sweep at full sensitivity. Eligibility is re-evaluated using corrected monthly count criteria, without the 25-day rule. MPC is median percentage change, positive for seizure-frequency reduction; bars are nonparametric 95% confidence intervals for the median. Dashed lines show perfect-detector MPC. Observed RTM is the study-specific trajectory indicator, not an isolated causal subtype.

How detector error can change apparent placebo response

Conceptual pathways, not a causal decomposition of the simulation

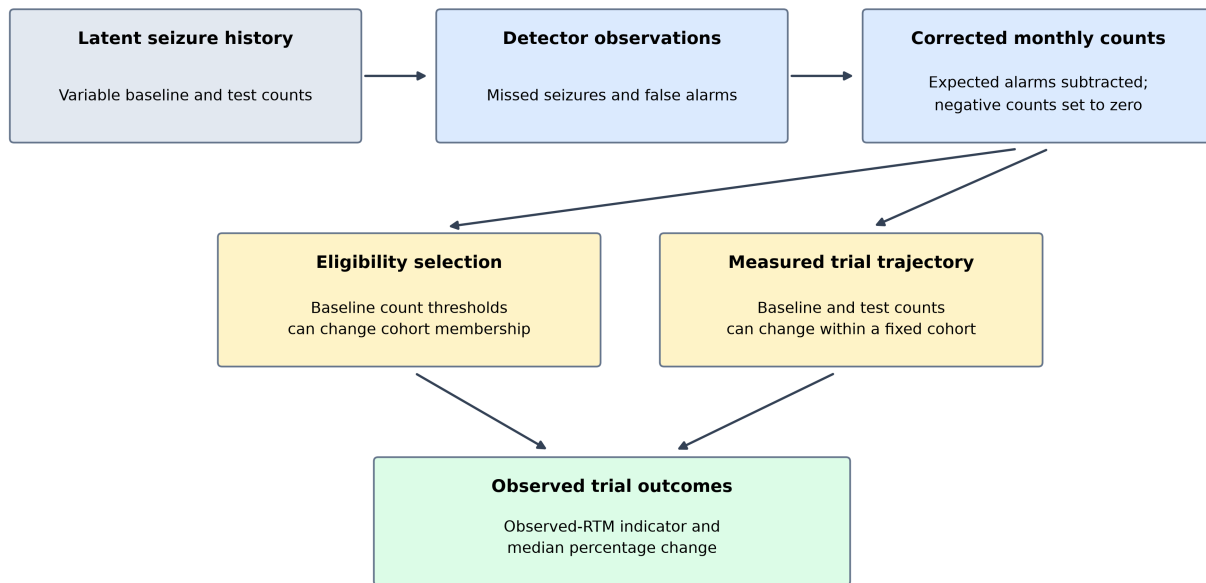


Figure 2. Conceptual pathways linking detector error to trial outcomes. Errors can change cohort membership through baseline eligibility and change measured baseline/test trajectories. Both affect observed RTM and percentage-change outcomes. The diagram does not quantify causal components; the Appendix compares fixed and reselected cohorts using shared latent histories and detector realizations.

Test Yourself

Choose the single correct answer for each question.

1. How can lower detector sensitivity affect enrollment when eligibility uses fixed measured seizure-count thresholds?

- A. It preserves the same eligible participants at every sensitivity.
- B. It removes the same fraction of eligible participants at every underlying seizure rate.
- C. It can favor higher underlying rates or unusually high measured baseline counts.
- D. It makes eligibility depend on the test-period seizure count.
- E. It lowers the numerical seizure-count thresholds automatically.

Answer: C. With fewer detected seizures, crossing a fixed threshold can favor higher underlying rates or unusually high measured baselines. A and B overlook this selection. D is incorrect because eligibility uses baseline counts. E is incorrect because the thresholds remain fixed.

2. What remains after subtracting the expected number of Poisson false alarms from a monthly seizure count?

- A. An exact count of the true seizures in that month.
- B. Random error from the difference between realized and expected false alarms.
- C. A detector sensitivity of 100%.
- D. The identity of each false alarm in that month.
- E. A guarantee that baseline eligibility is unaffected by false alarms.

Answer: B. Realized false alarms vary around their expectation, leaving that random difference after subtraction. A and D require knowing which events were false. The correction does not restore missed seizures, so C is incorrect. Residual error can change eligibility, making E incorrect. Setting negative corrected counts to zero can also introduce bias.

3. Which pattern meets this study's observed regression-to-the-mean definition?

- A. Any decrease in measured seizure frequency after baseline.
- B. A reduction of at least 50% from baseline.
- C. A difference between randomized drug and placebo groups.
- D. An improvement caused solely by detector error.
- E. Baseline exceeds the reference frequency and the test period is closer to that reference.

Answer: E. The indicator compares baseline and test frequencies with the detector-scale effective long-term reference. A and B do not require movement closer to that reference. C describes a treatment comparison. D is a causal attribution that the indicator does not establish.